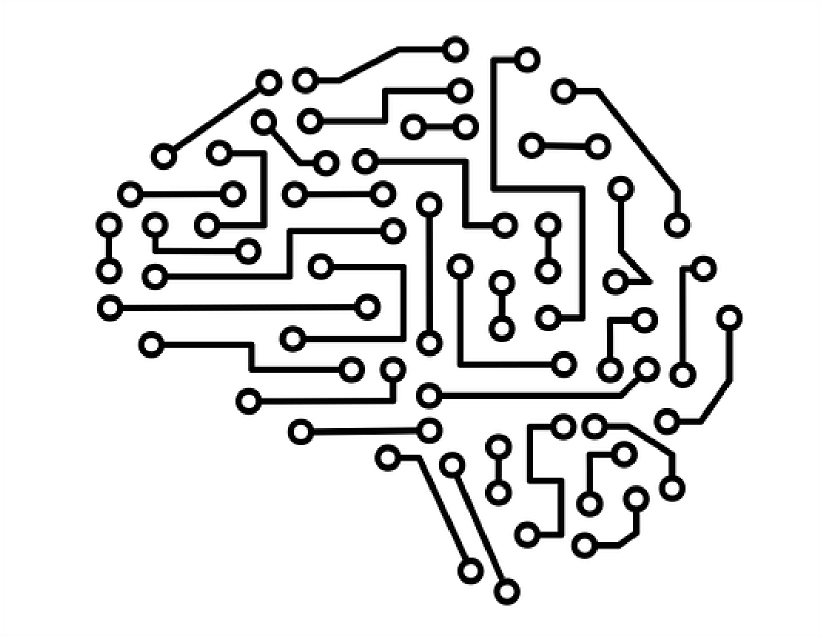
MACHINE LEARNING – PART 2

BY CHILL AND GROW

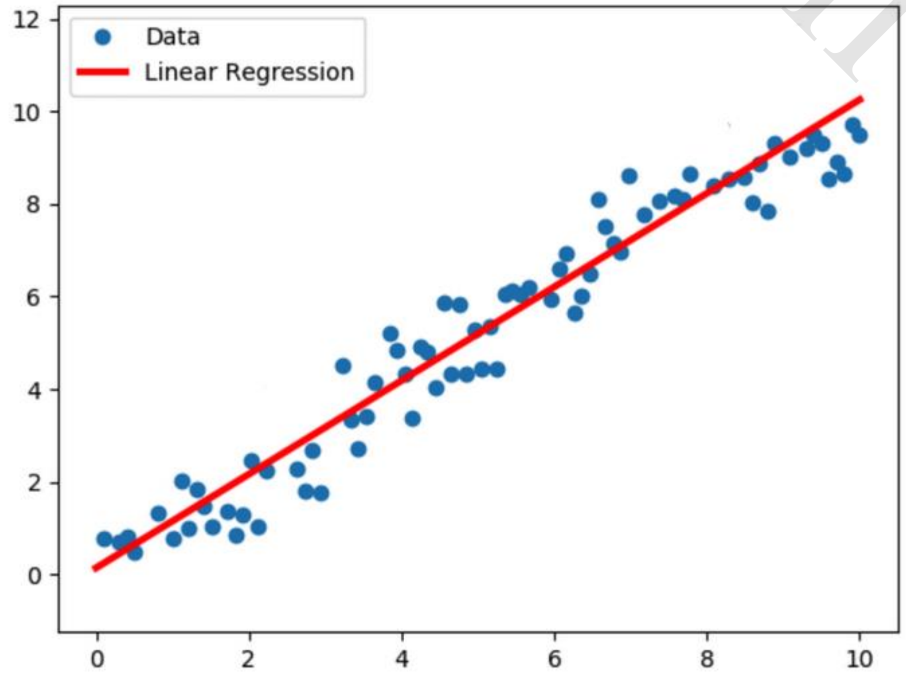


- Regression
- Types Of Regression

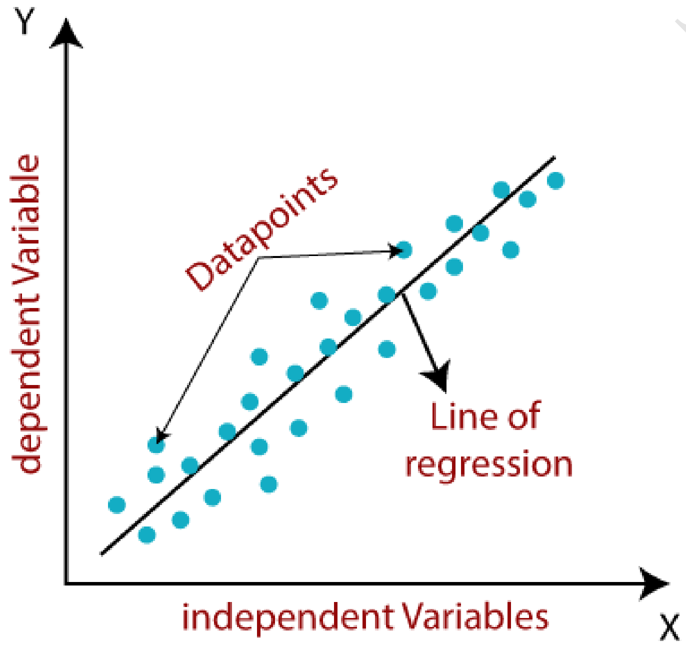
What is Regression?

Regression is a method to find the relationship between two or more variables. It helps us make predictions by drawing a straight line through data points. It's like having a ruler to estimate values based on known information.

LINEAR REGRESSION



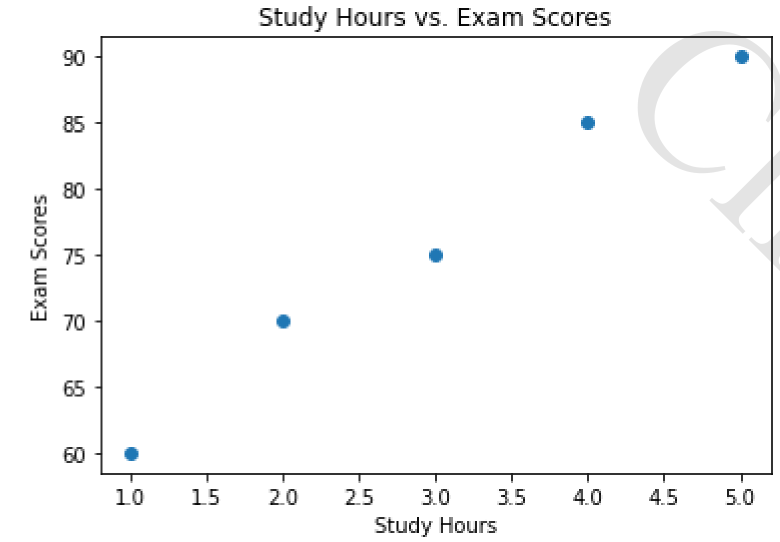
Linear regression is a method to find the best straight line that fits data points. It helps us understand how one variable changes with another. This line allows us to make predictions and see the overall trend in the data.

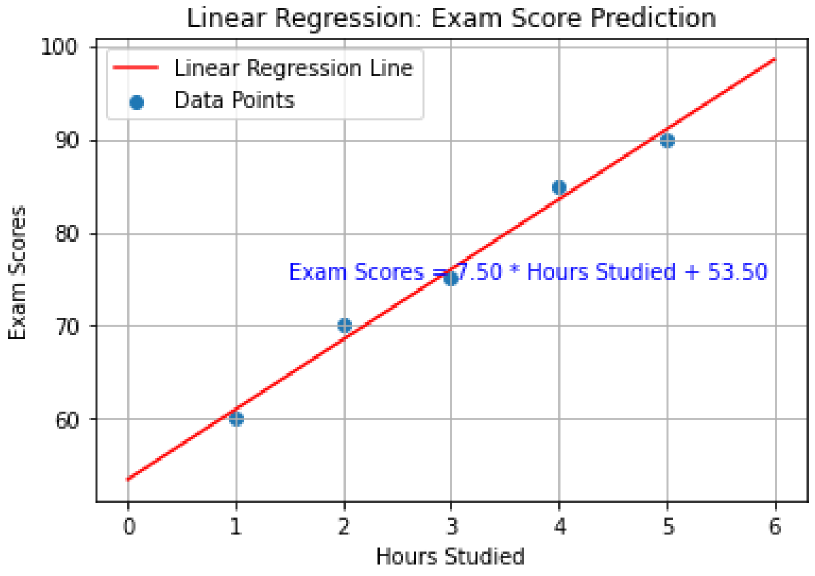


The line equation : y = mx + b

•"y" is the value we want to predict (dependent variable).
•"x" is the value we use to make the prediction (independent variable).
•"m" is the slope of the line, showing how much "y" changes for each unit change in "x."
•"b" is the y-intercept, where the line crosses the y-axis.

EXAMPLE



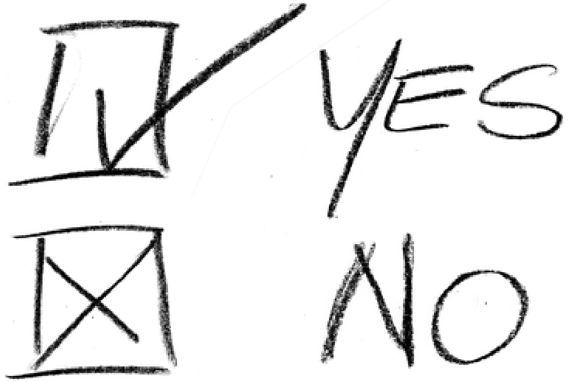


Study Hours (X)	Exam Scores (Y)
1	60
2	70
3	75
4	85
5	90

Y = mx + b.

With the line equation (Y = 10x + 50), we can make predictions. For example, if a student studies for 6 hours (X = 6), we can estimate their exam score by plugging the value into the equation: Y = 10 * 6 + 50 = 110.

LOGISTIC REGRESSION

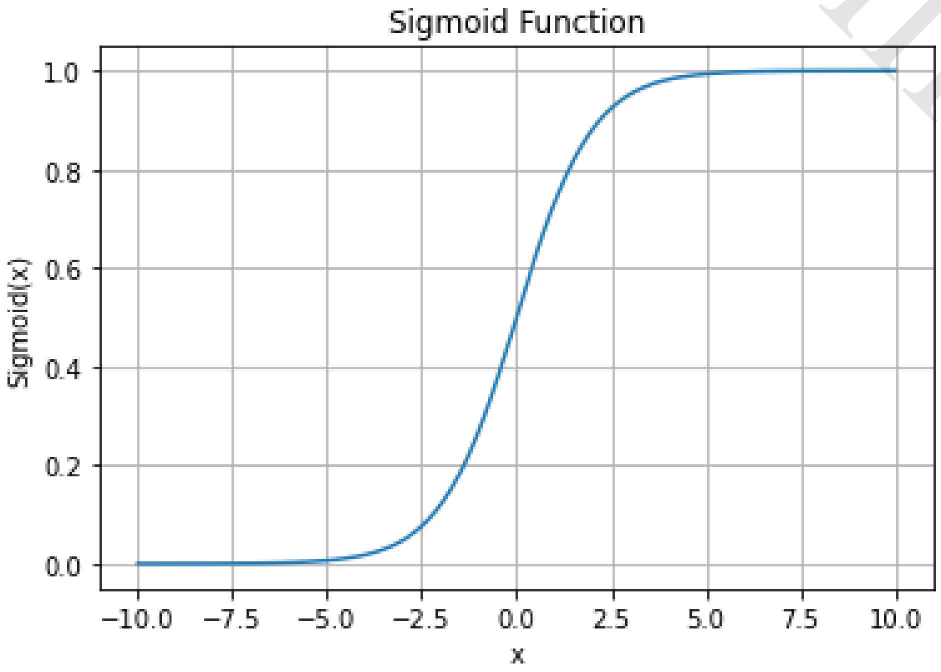


Logistic regression is a statistical technique used for binary classification tasks, where the goal is to predict one of two outcomes (e.g., yes or no, spam or not spam). It models the relationship between independent variables (features) and the probability of a binary outcome, using the sigmoid function to map the predictions to probabilities between 0 and 1. The result is a decision boundary that separates the two classes and allows us to make probabilistic predictions.

Example 1: Spam Email
Subject: "You've won a million dollars! Claim your prize now!"
Sender: randomprizegiveaway@example.com
Content: "Congratulations! You're a lucky winner! Click here to claim your million-dollar prize today!“

Example 2:
Not Spam Email Subject: "Weekly Newsletter: Latest Updates and Articles"
Sender: newsupdate@example.com
Content: "Dear subscriber, here are the latest news updates and articles for this week. Stay informed with our weekly newsletter!"

SIGMOID FUNCTION



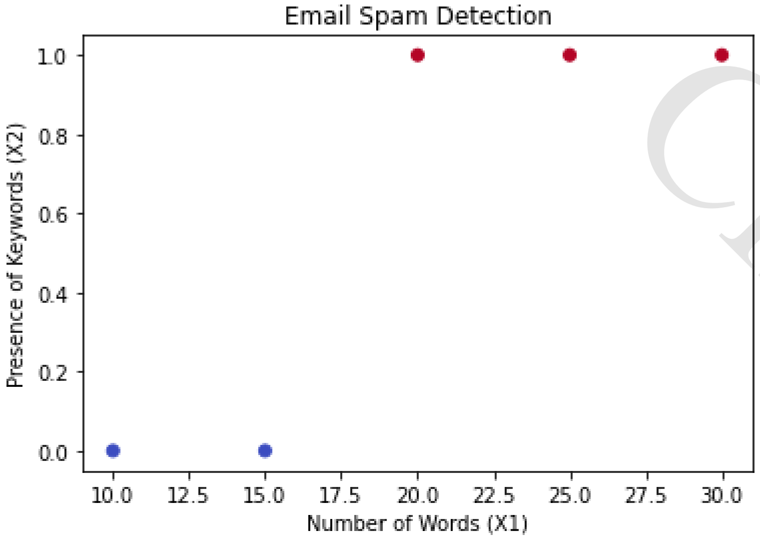
$y = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots + \beta n*Xn$

where:

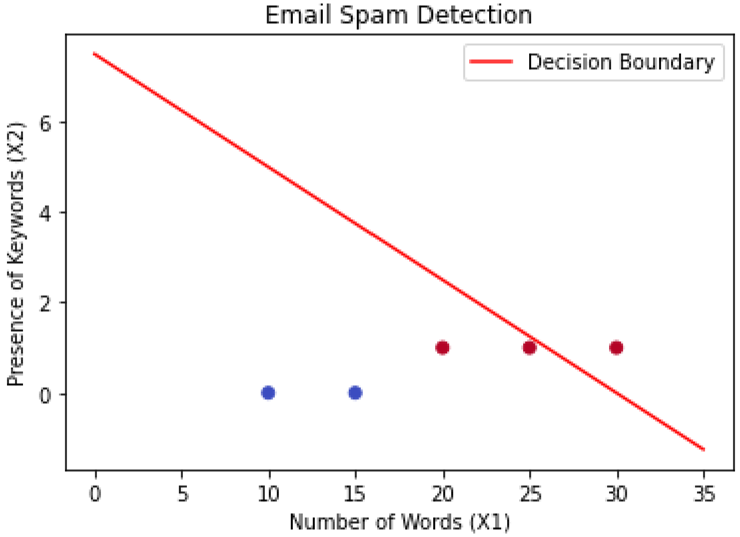
"β0", "β1", "β2", ..., "βn" are the coefficients of the linear regression model.
"X1", "X2", ..., "Xn" are the input features (e.g., number of words, presence of keywords, etc.).

Now, the sigmoid function comes into play. It takes the value “y" and converts it into a probability value between 0 and 1. The formula for the sigmoid function is:

$sigmoid(y) = 1 / (1 + e^{(-y)})$



Number of Words (X1)	Presence of Keywords (X2)	Is Spam (Label)
25	1	1
15	0	0
30	1	1
10	0	0
20	1	1



$$\text{Probability of Spam} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n)}}$$